

Comparison of Supervised Machine Learning Algorithms for Fake News Detection

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Abstract

The proliferation of fake news on digital platforms poses significant risks to the integrity of information dissemination and public discourse. It provides harmful or false data to many people for reasons ranging from the nefarious to the benign. This study applies supervised machine learning to detect fake news using the “Fake and Real News” dataset from Kaggle. Articles are classified as fake or real using semantic, linguistic, and metadata features after artifact-aware preprocessing that removes publisher boilerplate, outlet identifiers, and near-duplicates to reduce spurious cues. Three primary models are compared for effectiveness in distinguishing true from false information: Naive Bayes, penalized logistic regression using GLMNET, and Decision Trees. A Support Vector Machine is included as a supplementary linear benchmark. Model performance is evaluated using accuracy, precision, recall, F1-score, and the area under the ROC curve to provide a comprehensive assessment of classification capability. The findings are intended to enhance reliable news consumption, foster informed decision making, and identify approaches that can help mitigate the spread of misinformation online.

Keywords: Fake news detection, machine learning, data mining, misinformation, supervised learning, news classification

1. Introduction

The rise of digital platforms has transformed the way information is created, shared, and consumed. While these platforms enable rapid access to news, they also provide fertile ground for the spread of misinformation. Fake news has emerged as a critical problem because it reduces public trust in mainstream media across political divides and can distort civic engagement (Ognyanova, Lazer, Robertson, & Wilson, 2020). Traditional fact-checking organizations play an important role but cannot scale effectively against the volume and velocity of false information online.

Supervised machine learning provides a scalable approach to detecting fake news by leveraging linguistic, semantic, and metadata features to classify articles as real or fake. By applying these methods to labeled datasets, researchers can evaluate model effectiveness in distinguishing credible journalism from misinformation. This study uses the Kaggle

“Fake and Real News” dataset to compare supervised machine learning algorithms and assess their performance with multiple evaluation metrics. The aim is to identify effective classification strategies that can support reliable news consumption and help mitigate the harmful effects of misinformation in digital spaces.

2. Problem Statement

The spread of fake news on digital platforms threatens the reliability of online information and public trust. This study addresses the problem of automatically distinguishing real from fake news articles. The objective is to apply supervised machine learning models for classification, with success measured by

accuracy, precision, recall, F1-score, and area under the ROC curve (AUC).

3. Literature Review

Fake news has been an ongoing problem for years, causing extensive research into potential solutions as well as obstacles with prediction.

According to Murayama (2021), fact verification is an important part of fake news detection and publicly available datasets have been used for years to assess the truthfulness of claims. In recent years, however, fake news has grown more complex, and detection comes with new challenges. For example, news articles from unreliable sources are considered fake news but do not necessarily have the characteristics of fake news. Recent fake news detection models include detection of context such as news publishers, user information, temporal information, and network information.

Evaluation metrics for fake news detection typically include precision, recall, F1-score, and AUC in addition to accuracy, to address possible skewed datasets (Shu et al., 2017). While our dataset is balanced, we still use these evaluation metrics for consistency with other studies and to get a full picture of the model's performance.

Proper preprocessing is necessary for ensuring that models learn linguistic patterns rather than artifacts of time and context. Murayama, Wakamiya, and Aramaki (2021) studied difficulties with bias in fake news detection as fake news detection datasets often depend on a specific time period. Since topics and content of news change drastically over time, fake news detection models built on these datasets often have deceptively high accuracy and need preprocessing techniques to reduce bias. These techniques include Named Entity Recognition (NER) to mask proper nouns by replacing named entities with their NER tags, replacing people's names with their occupation, or even removing named entities completely.

Khan et al. (2021) conducted a benchmark study of machine learning models for online fake news detection. They compared the performance of traditional machine learning models, deep learning models, and advanced pre-trained language models. Out of the traditional machine learning models, Naive Bayes with n-gram stood out as it reached a

0.93 accuracy and 0.93 F1-score in the combined corpus. For comparison, the Decision Tree model achieved only a 0.67 accuracy and 0.70 F1-score. The study found that deep learning models outperformed most traditional learning models, though Naive Bayes with n-grams achieved comparable results. Advanced pre-trained language models outperformed both traditional and deep learning models, with RoBERTa achieving 0.96 accuracy and F1-score. While deep learning models and large language models are outside the scope of our study, these results show the advancements moving the field forward.

3. Dataset Selection

This study uses the *Fake and Real News* dataset hosted on Kaggle (Bisaillon, 2019). The dataset contains two CSV files: *True.csv* with authentic news articles and *Fake.csv* with fabricated news articles. Each file includes article text, headlines, publication dates, and source information, providing sufficient depth for linguistic and metadata analysis. Kaggle was chosen as the data source because it is widely used in academic research, offers reproducibility, and provides a clear distinction between real and fake labels that supports supervised learning tasks.

The dataset was collected from a variety of online outlets, though the precise curation methods may introduce some limitations. For example, differences in publisher style and formatting can act as artifacts that models exploit rather than focusing solely on semantic content. To address this, artifact-aware preprocessing was applied, including the removal of publisher identifiers and boilerplate text. Despite these limitations, the dataset remains a well-recognized benchmark for fake news detection and is suitable for evaluating the performance of machine learning models in this study.

4. Data Cleaning

The *Fake and Real News* dataset from Kaggle contained two CSV files: *True.csv* and *Fake.csv*. Each dataset included article text, headlines, publication dates, and source information. Before training any models, the data was cleaned in R using the *tm* package's text preprocessing functions.

4.1 Data Labeling and Splitting

A label column was added to identify "REAL" and "FAKE" instances, ensuring that each article could be clearly categorized for supervised learning. The two datasets were then merged into a single dataframe to streamline analysis and model training. To preserve class balance and avoid bias toward either category, the data was stratified into an 80% training set and a 20% testing set. Stratification maintained the proportion of real and fake articles in both subsets, which is critical for building fair models that generalize beyond the training data.

To prevent information leakage between training and testing, separate corpora were created for each split. This design decision ensured that preprocessing transformations, such as vocabulary construction, were based solely on the training set before being applied to the testing

set. Without this safeguard, the models could inadvertently learn from data they were later evaluated on, inflating accuracy in unrealistic ways.

4.2 Text Preprocessing

The raw dataset contained a variety of noise that could obscure meaningful linguistic signals. Preprocessing was therefore applied to standardize the text into a consistent, machine-readable form. These steps are widely recognized as essential in natural language processing pipelines for text classification tasks (Chai, 2023; Alqadi, 2025). The following steps were performed:

- Converting all text to lowercase to ensure consistent token matching.
- Removing hyperlinks, Twitter handles, punctuation, and numbers to reduce noise.
- Removing English stopwords (e.g., "the," "and") as well as dataset-specific terms ("reuter," "reuters") that offered little predictive value.
- Stripping extra whitespace and applying Porter stemming to reduce words to their root form.

```
FAKE REAL
18785 17134
Class balance – TEST:

FAKE REAL
4696 4283

Top terms by document frequency (post-trim):
term doc_freq
1 said 26051
2 presid 18772
3 will 18083
4 say 17702
5 trump 17555
6 reuter 17230
7 state 17203
8 one 16053
9 peopl 15194
10 also 14947
11 donald 14606
12 year 13958
13 like 13278
14 time 13028
15 new 12887
```

Figure 1. Distribution of fake vs. real articles in the training and testing splits, along with the top fifteen terms by document frequency after preprocessing.

4.3 Normalization and Stemming

Beyond cleaning, normalization techniques were applied to further refine the corpus. Extra whitespace and formatting inconsistencies were

stripped out to produce uniformly structured text. More importantly, **Porter stemming** was implemented to reduce words to their morphological root forms. For example, "running," "runs," and "ran" would all be reduced to "run."

Stemming helped reduce vocabulary size and grouped together words with shared meaning, which allowed the models to focus on semantic signals rather than surface-level variations. This step is particularly important in text mining, as sparse vocabularies can dilute statistical power and increase computational complexity. By consolidating similar tokens, the stemming process improved efficiency and reinforced consistency across the dataset (Chai, 2023).

5. Data Exploration and Analysis

Exploratory data analysis (EDA) was conducted to better understand the dataset and to highlight differences between real and fake articles before model training. This stage was important for identifying patterns in metadata as well as the underlying text that could guide feature engineering and model selection

5.1 Metadata Features

Basic descriptive statistics such as character count, word count, and punctuation use were calculated for each article. These features showed clear stylistic contrasts between the two classes. Fake articles tended to be shorter on average and contained more exclamation marks, which often signal attempts to convey urgency or emotional emphasis. Real articles were generally longer and more formal in tone. These differences in metadata align with prior studies showing that fake news relies more heavily on stylistic cues than on substantive reporting (Reddy, 2020). Boxplots and histograms were used to visualize these patterns and confirm their consistency across the dataset.

5.2 Token Distribution Analysis

A closer look at token distributions revealed overlapping but distinct linguistic trends between real and fake news. Common words such as "said," "trump," and "government" appeared across both classes, but fake news articles contained more emotionally charged or politically biased vocabulary. In contrast, real news leaned toward institutional and factual language. This finding matches observations in the literature that word frequency and lexical choice are key indicators for deception detection (Oshikawa et al., 2018). Word clouds and frequency charts

were generated to highlight these differences and illustrate how vocabulary can signal article type.

5.3 Vocabulary Pruning and Selection

To reduce dimensionality and improve computational efficiency, sparse terms were removed, and the vocabulary was limited to the 1,200 most frequent terms. This pruning ensured that infrequent and noisy tokens were excluded while still preserving words with the greatest predictive value. By narrowing the vocabulary in this way, the dataset remained representative while avoiding the sparsity that can weaken statistical modeling.

6. Machine Learning Method Selection and Model Training

Three supervised machine learning models were selected for comparison:

1. **Naive Bayes** (Bernoulli variant) – chosen for its simplicity, speed, and strong baseline performance in text classification. Binary features (term presence/absence) were used for stability.
2. **GLMNET** (penalized logistic regression with L1 regularization) – selected for its ability to handle high-dimensional sparse data and perform feature selection automatically. TF-IDF weighted features were used.
3. **Decision Tree (rpart)** – chosen for interpretability and rule extraction capability, though not expected to outperform linear models in sparse text settings.

The train-test split was **80/20 stratified** by class label. For GLMNET, a 3-fold cross-validation was performed to select the regularization parameter (λ). Class imbalance was addressed with inverse-frequency sample weighting. For Naive Bayes, Laplace smoothing was set to 1 to prevent zero-probability issues. For the Decision Tree, cross-validation determined the complexity parameter (cp) to avoid overfitting.

Metadata features were appended to text-based term matrices to potentially capture stylistic differences not visible in token counts alone.

6.1 Naive Bayes

Naive Bayes was implemented as a baseline classifier due to its simplicity, efficiency, and strong performance in text classification tasks. The Bernoulli variant was selected, which models features as binary indicators of term presence or absence. This approach reduces instability when working with sparse, high-dimensional data.

Laplace smoothing ($\alpha = 1$) was applied to prevent zero-probability issues for unseen words.

6.2 GLMNET

GLMNET was applied as a penalized logistic regression model with L1 (lasso) regularization. This method is well suited for text mining because it can handle high-dimensional sparse data while automatically performing feature selection by shrinking irrelevant coefficients to zero. Term frequency-inverse document frequency (TF-IDF) weighting was used to represent article text, capturing word importance relative to the corpus. A 3-fold cross-validation procedure was used to select the regularization parameter λ , and class imbalance was mitigated through inverse-frequency sample weighting.

6.3 Decision Trees

Decision Trees were trained using the CART algorithm implemented in the rpart package. Splits were based on features that minimized the Gini impurity at each node. Metadata such as word count and punctuation frequency were included alongside token features to improve interpretability. Cross-validation was used to select the complexity parameter (cp) and avoid overfitting.

6.4 Support Vector Machine

A linear Support Vector Machine (SVM) was trained as a supplementary benchmark. SVMs are margin-based classifiers that maximize the separation between real and fake news in high-dimensional space, making them effective for text classification. The linear kernel was chosen for efficiency and compatibility with sparse TF-IDF representations. Hyperparameters were tuned using cross-validation, and class imbalance was addressed with balanced class weights.

7. Model Evaluation

Models were evaluated on the held-out test set using **accuracy**, **precision**, **recall**, **F1-score**, and **AUC** (area under the ROC curve), with "FAKE" treated as the positive class.

Model	Accuracy	Precision	Recall	F1	AUC
GLMNET	0.981	0.981	0.983	0.982	0.997
Naive Bayes	0.943	0.955	0.935	0.945	0.974
Decision Tree	0.876	0.871	0.895	0.883	0.903

GLMNET achieved the highest balanced performance (F1-score) and AUC, suggesting that penalized logistic regression with TF-IDF weighting effectively captured discriminative patterns in the text. Its strong metrics, including 98.1% accuracy, F1-score of 0.982, and an AUC of 0.997, highlight its ability to balance predictive power and generalization.

Despite its simplicity, Naive Bayes achieved strong results with an accuracy of 94.3%, F1-score of 0.945, and AUC of 0.975, demonstrating its effectiveness as a lightweight fake news detector.

While Decision Trees offer clear interpretability through if-then rules, their predictive strength was lower compared to linear models. The model achieved 87.6% accuracy, F1-score of 0.883, and AUC of 0.902, confirming that while Decision Trees provide transparency, they are less competitive on sparse, high-dimensional text data.

The SVM achieved results comparable to GLMNET, with high accuracy and strong F1-score, further reinforcing the effectiveness of linear margin-based models for fake news detection. However, the computational cost of SVM training was higher than GLMNET, making penalized logistic regression the more efficient choice.

Confusion matrix analysis revealed that misclassifications were relatively balanced between false positives and false negatives. Visualizations of the decision tree highlighted certain token and metadata splits, such as word count thresholds, as decision boundaries.

8. Discussion

The evaluation showed clear differences in model performance. GLMNET (penalized logistic regression) achieved the highest F1-score and AUC, demonstrating strength in handling sparse, high-dimensional text through TF-IDF weighting and regularization. Naive Bayes performed slightly below GLMNET but provided strong recall, making it valuable where catching more fake articles is prioritized. Decision Trees lagged in predictive power, highlighting their weakness with sparse data despite offering interpretability. The supplementary Support Vector Machine (SVM) produced balanced results close to GLMNET, reinforcing the value of linear classifiers, though at higher computational cost.

An important issue emerged during preprocessing: the need to remove publisher identifiers such as "Reuters." Without this,

models relied on dataset artifacts rather than genuine linguistic cues, artificially boosting performance. Once removed, results reflected true classification ability, underscoring the importance of artifact-aware preprocessing. Largely, linear models with regularization or margin-based optimization proved best suited for fake news detection, while Decision Trees remain useful when interpretability is prioritized over predictive strength.

9. Limitations

This study is subject to several limitations. First, the dataset is English-only and sourced from a limited set of publishers, which restricts the generalizability of the findings to multilingual or social media-based misinformation. Second, although artifact-aware preprocessing was applied, subtle dataset biases may remain, such as writing style differences linked to outlets. Third, the models were evaluated on a static dataset with balanced splits, whereas real-world news streams are dynamic and highly imbalanced toward authentic content. Finally, while the inclusion of a Support Vector Machine provided an additional linear benchmark, the study did not explore ensemble methods or neural architectures that may outperform traditional supervised models.

10. Future Work

Future research should extend this work in three directions. First, datasets should be diversified to include multilingual news sources, social media posts, and fact-checker annotations to better capture real-world misinformation contexts. Second, advanced models such as deep learning

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(e.g., recurrent neural networks, transformers) could be tested to evaluate whether they surpass linear models in detecting nuanced rhetorical or semantic patterns. Third, practical applications of fake news detection require streaming models capable of real-time classification, continual learning, and robustness against adversarial manipulation. Finally, interpretability remains essential; hybrid approaches that combine the predictive strength of linear or deep models with the transparency of rule-based methods may provide a balanced path forward.

11. Conclusion

This study compared supervised machine learning models for detecting fake news using a labeled dataset of real and fabricated articles. Among the tested models, GLMNET and SVM showed the strongest overall performance, with Naive Bayes offering a strong lightweight alternative and Decision Trees providing interpretability at the cost of accuracy. Preprocessing choices such as removing publisher identifiers like "Reuters" proved essential to ensure that models relied on genuine linguistic and semantic cues rather than artifacts. The results suggest that linear models with regularization or margin-based optimization are the most reliable for this task, while interpretable models remain useful for transparency. By combining effective preprocessing with strong classification methods, machine learning can provide valuable tools to reduce the spread of misinformation and support trustworthy news consumption.

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